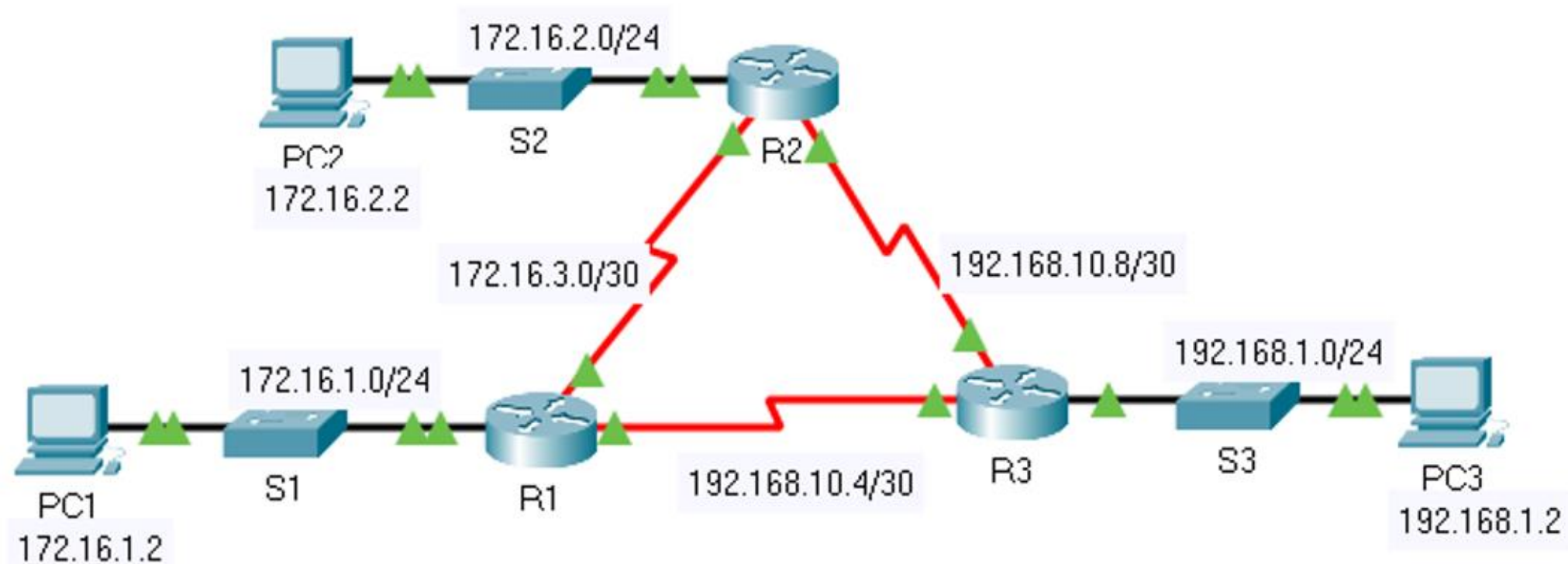


Lab 2.2 - Configuring OSPFv2 in a Single Area

- Part 1: Configure OSPFv2 Routing
- Part 2: Verify the Configurations

Network Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	N/A
	S0/0/1	192.168.10.5	255.255.255.252	N/A
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	N/A
	S0/0/1	192.168.10.9	255.255.255.252	N/A
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	N/A
	S0/0/1	192.168.10.10	255.255.255.252	N/A
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1

Background

- In this activity, the IP addressing is already configured.
- You are responsible for configuring the three router topology with basic single area OSPFv2 and then verifying connectivity between end devices.

Part 1: Configure OSPFv2 Routing

Step 1: Configure OSPF on the R1, R2 and R3.

Use the following requirements to configure OSPF routing on all three routers:

- **Process ID 10**
- **Router ID for each router: R1 = 1.1.1.1; R2 = 2.2.2.2; R3 = 3.3.3.3**
- **Network address for each interface**
- **LAN interface set to passive (do not use the default keyword)**

Step 2: Verify OSPF routing is operational.

- **On each router, the routing table should now have a route to every network in the topology.**

Part 2: Verify the Configurations

- Each PC should be able to ping the other two PCs. If not, check your configurations.

Configure OSPFv2 Routing

Router R1

```
router ospf 10
router-id 1.1.1.1
network 172.16.1.0 0.0.0.255 area 0
network 172.16.3.0 0.0.0.3 area 0
network 192.168.10.4 0.0.0.3 area 0
passive-interface GigabitEthernet0/0
end
```

Configure OSPFv2 Routing

Router R2

```
router ospf 10
router-id 2.2.2.2
network 172.16.2.0 0.0.0.255 area 0
network 172.16.3.0 0.0.0.3 area 0
network 192.168.10.8 0.0.0.3 area 0
passive-interface GigabitEthernet0/0
end
```

Configure OSPFv2 Routing

Router R3

```
router ospf 10  
router-id 3.3.3.3  
network 192.168.1.0 0.0.0.255 area 0  
network 192.168.10.4 0.0.0.3 area 0  
network 192.168.10.8 0.0.0.3 area 0  
passive-interface GigabitEthernet0/0  
end
```

Verify the Configurations

```
R1#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.16.1.1 0.0.0.0 area 0
    192.168.10.5 0.0.0.0 area 0
    172.16.3.1 0.0.0.0 area 0
  Passive Interface(s):
    GigabitEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:00:03
    2.2.2.2          110          00:00:02
    3.3.3.3          110          00:00:03
  Distance: (default is 110)
```

Verify the Configurations

```
R2#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.16.2.0 0.0.0.255 area 0
    172.16.3.0 0.0.0.3 area 0
    192.168.10.8 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:01:01
    2.2.2.2          110          00:01:02
    3.3.3.3          110          00:01:01
  Distance: (default is 110)
```

Verify the Configurations

```
R3#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.8 0.0.0.3 area 0
    192.168.10.4 0.0.0.3 area 0
    192.168.1.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:01:39
    2.2.2.2          110          00:01:39
    3.3.3.3          110          00:01:39
  Distance: (default is 110)
```

Verify the Configurations

PC1

Physical Config Desktop Programming

Command Prompt

```

Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 15ms, Average = 4ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=14ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 4ms

C:\>ping 172.16.2.2

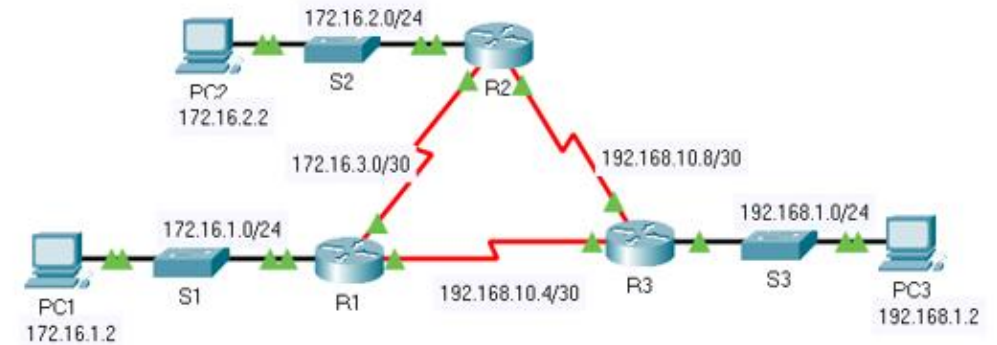
Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=19ms TTL=126
Reply from 172.16.2.2: bytes=32 time=15ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 19ms, Average = 9ms

C:\>

```



Verify the Configurations

PC2

Physical Config Desktop Programming

Command Prompt

```

Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=18ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 18ms, Average = 5ms

C:\>ping 192.168.1.2

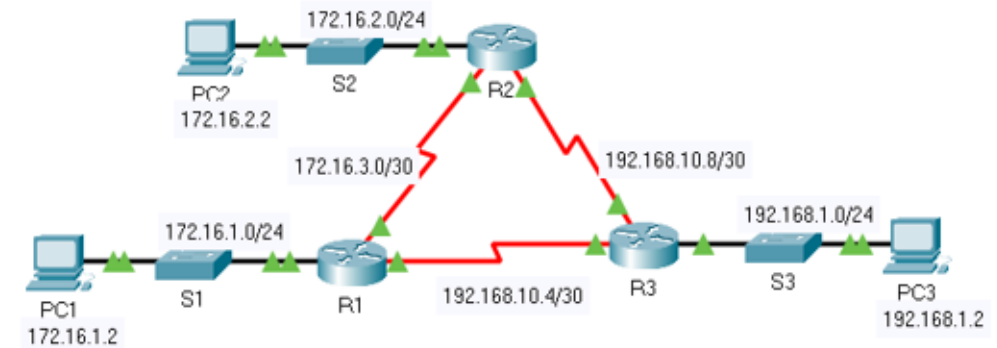
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=12ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 3ms

C:\>

```



Verify the Configurations

PC3

Physical Config Desktop Programming

Command Prompt

```

Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=16ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 4ms

C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=18ms TTL=126
Reply from 172.16.2.2: bytes=32 time=2ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126
Reply from 172.16.2.2: bytes=32 time=2ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 18ms, Average = 5ms

C:\>

```

